Figure 10.1 In a microservices architecture a single application is broken into loosely coupled services that communicate using remote APIs. In this example, a fictional web store has an API for web clients that calls to internal services to check stock levels, process payments, and arrange shipping when an order is placed.

10.1 Microservice APIs on Kubernetes

Although the concepts in this chapter are applicable to most microservice deployments, in recent years the Kubernetes project (https://kubernetes.io) has emerged as a leading approach to deploying and managing microservices in production. To keep things concrete, you'll use Kubernetes to deploy the examples in this part of the book. Appendix B has detailed instructions on how to set up the Minikube environment for running Kubernetes on your development machine. You should follow those instructions now before continuing with the chapter.

The basic concepts of Kubernetes relevant to deploying an API are shown in figure 10.2. A Kubernetes cluster consists of a set of *nodes*, which are either physical or virtual machines (VMs) running the Kubernetes software. When you deploy an app to the cluster, Kubernetes replicates the app across nodes to achieve availability and scalability requirements that you specify. For example, you might specify that you always require at least three copies of your app to be running, so that if one fails the other two can handle the load. Kubernetes takes care of ensuring these